

# MDIE

## Modification-Disentangled Identity Encoder

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A training-only framework that makes face recognition stable across masks, glasses, surgery, age, occlusion, low light, blur, and adversarial perturbation.

**Mrinal**

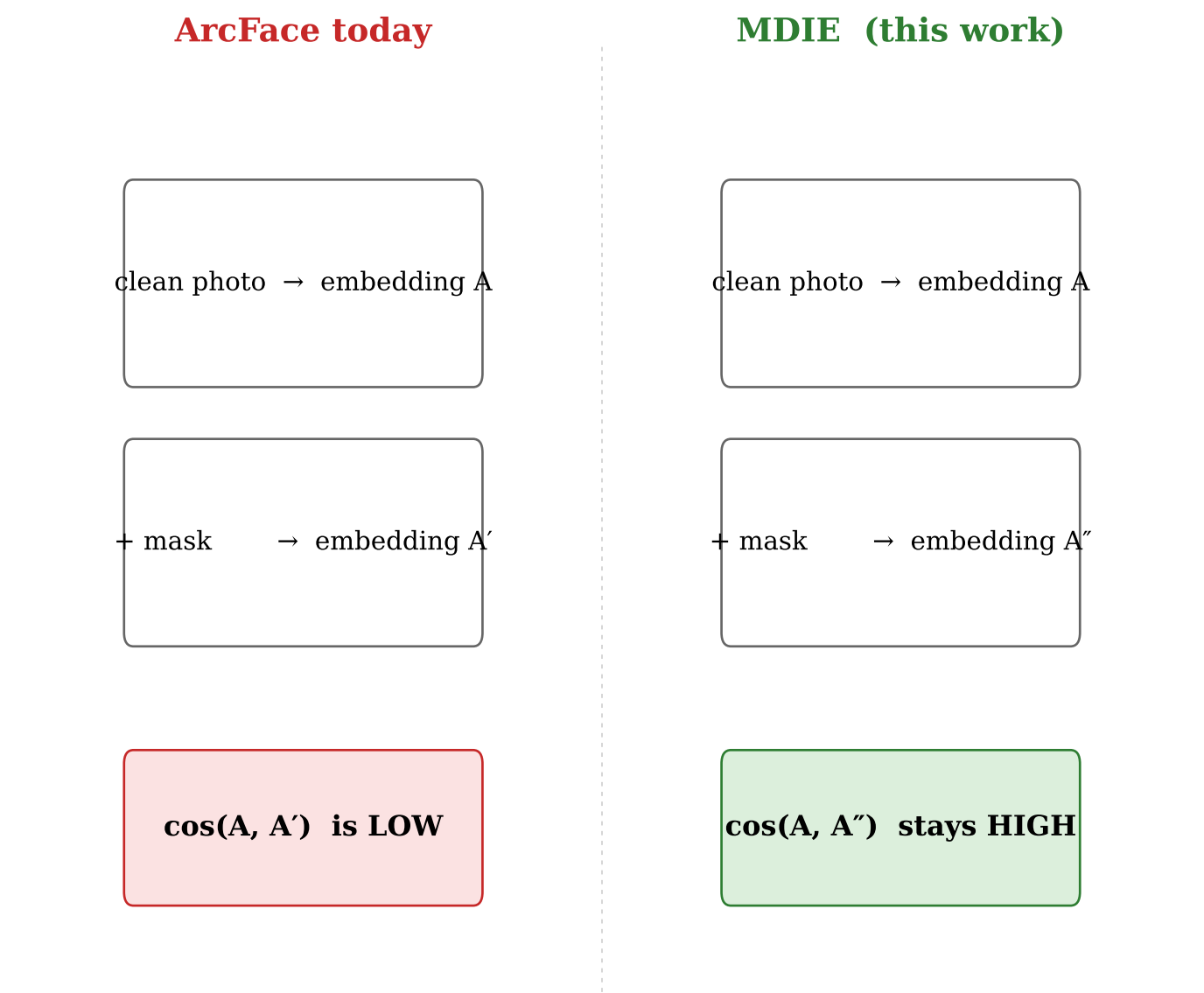
*Visual explainer*

*what we built · why it is novel · how it works*

# The problem

State-of-the-art face encoders collapse under image modifications.

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Across nine modifications on LFW, baseline AUC drops by 0.13 to 0.19.  
An early MDIE smoke run already shows a drop of only 0.01.

# Nine modifications we stress-test

Generated on the fly from clean LFW images.

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**surgery\_nose**

**surgery\_jaw**

**disguise\_glasses**

**disguise\_mask**

**occlusion\_random**

**aging**

**low\_light**

**motion\_blur**

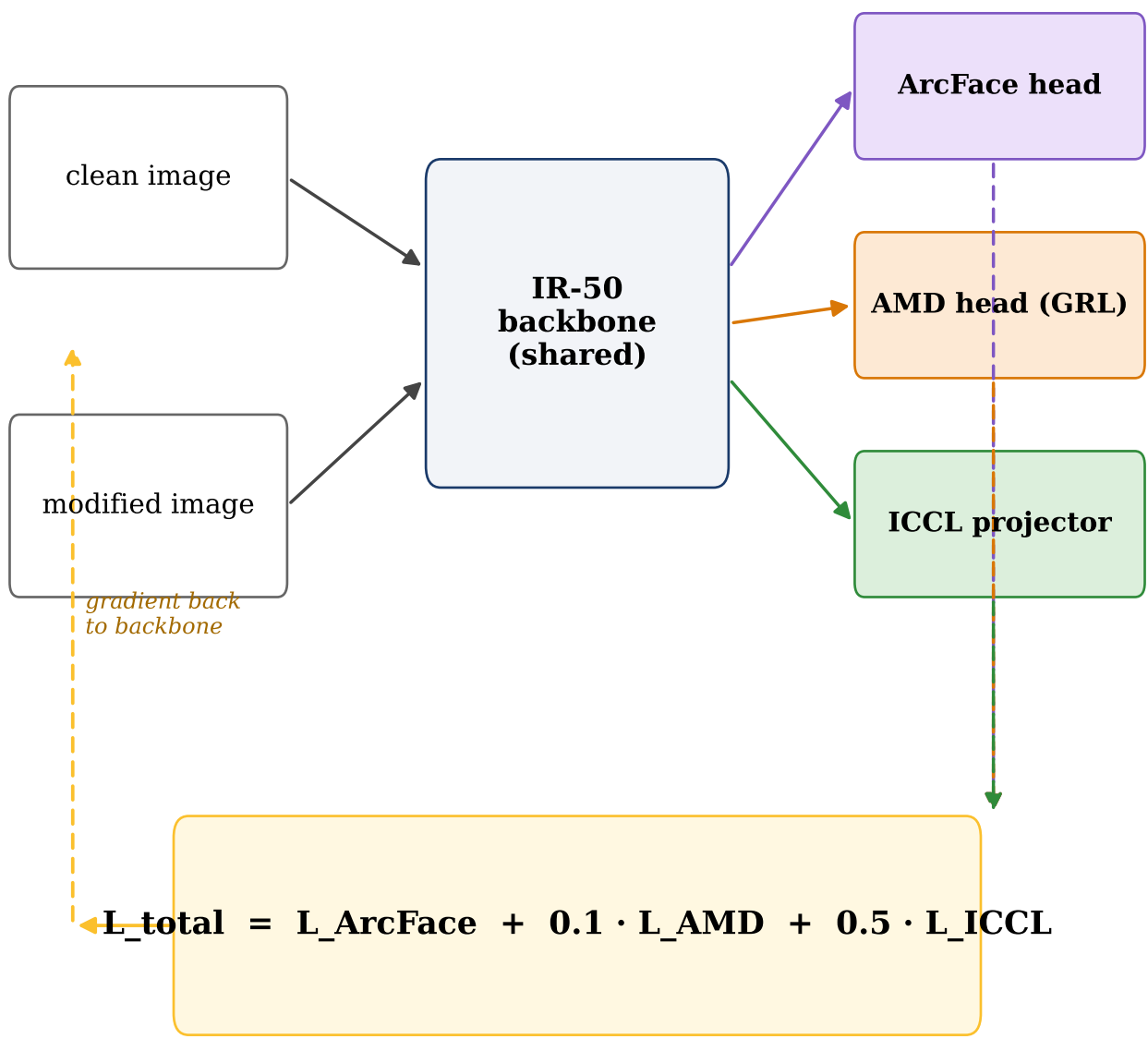
**adversarial**

Synthetic so we get unlimited paired (clean, modified) samples.

Real benchmarks (MFR2, CALFW, AgeDB-30, IIITD-Surgery, IJB-C) are the test set.

# How MDIE trains

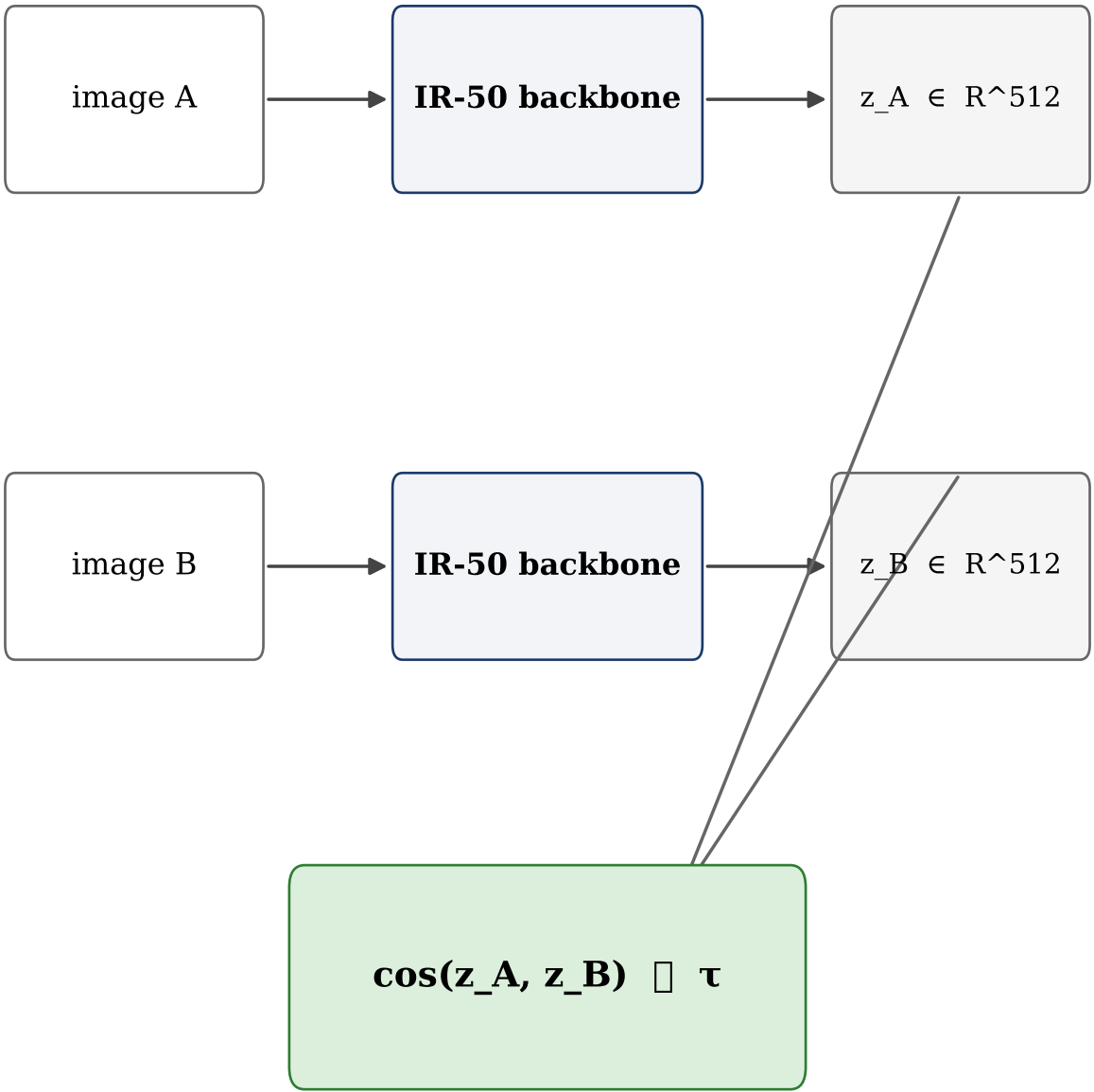
One backbone, three heads, one composite loss.



*All three heads are discarded at inference. Only the backbone ships.*

# How MDIE runs at inference

Same architecture as ArcFace. Zero extra parameters, zero extra latency.



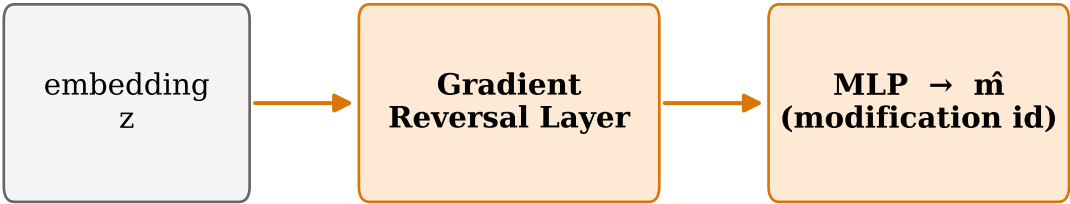
**Inference parameter count      ArcFace : 43.91 M      MDIE : 43.91 M**

*AMD head + ICCL projector exist only during training.*

# Novelty 1 · AMD

Adversarial Modification Disentanglement.

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## What the GRL does

forward pass    →    identity (pass-through)  
backward pass   →   multiply gradient by  $-\lambda$

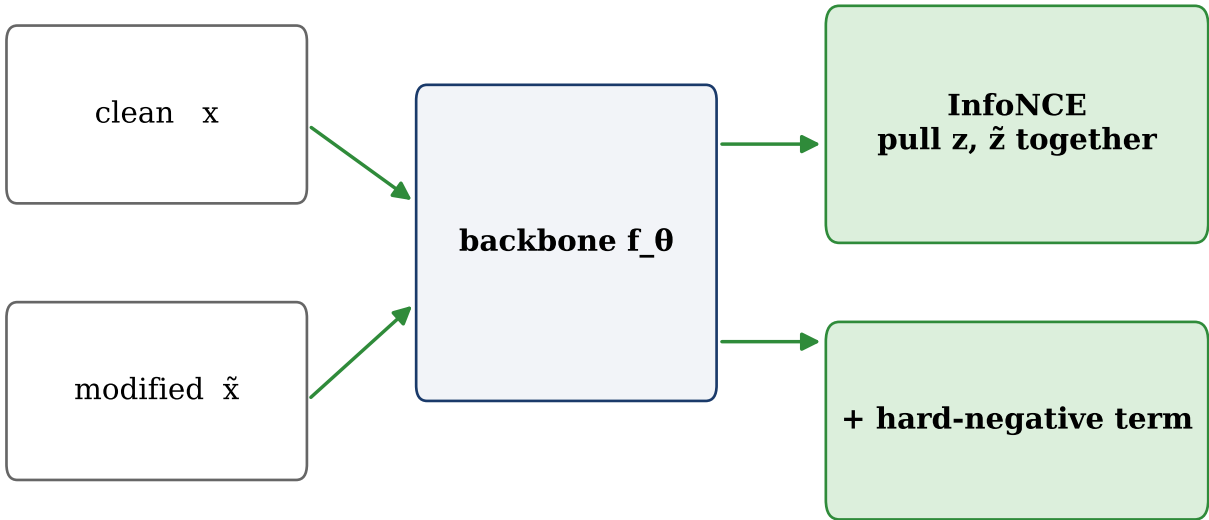
$$\min_{\theta} \max_{\varphi} -L_{AMD}(\theta, \varphi) \quad \Rightarrow \quad I(z ; \text{modification}) \rightarrow 0$$

The backbone is incentivised to make the modification UNRECOVERABLE from the embedding. Strictly stronger than data augmentation.

*Inference cost: 0 parameters, 0 FLOPs.*

# Novelty 2 · ICCL

Identity-Consistency Contrastive Loss.



## The hard-negative idea

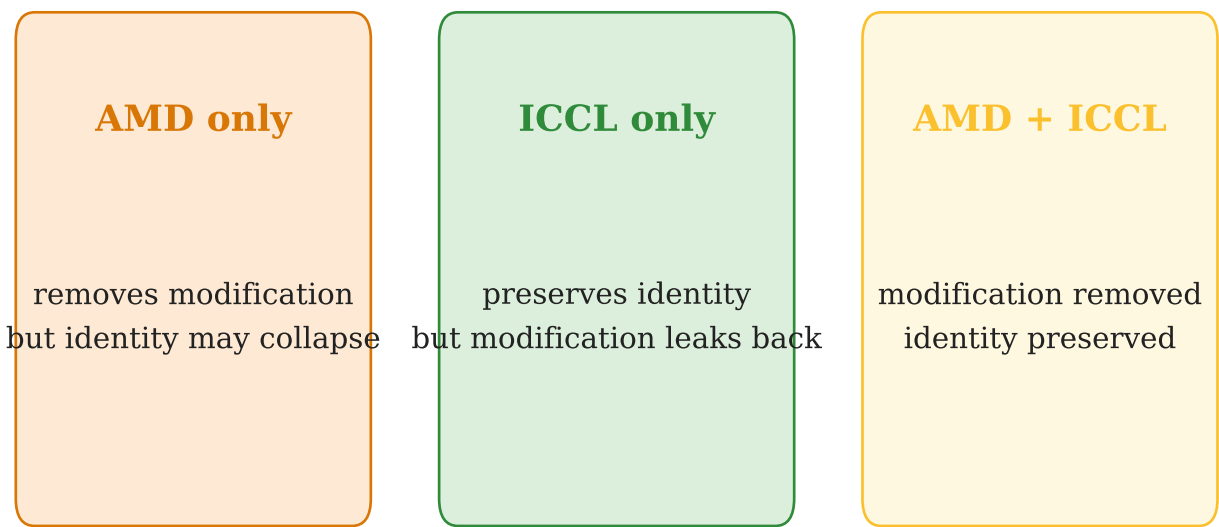
different person + SAME modification = the case that fools the model

$$\mathbf{L}_{\text{hard}} = \text{mean of } \max(0, \cos(\mathbf{z}_i, \tilde{\mathbf{z}}_j) - 0.3) \cdot \mathbf{1}[\mathbf{m}_i = \mathbf{m}_j, \mathbf{y}_i \neq \mathbf{y}_j]$$

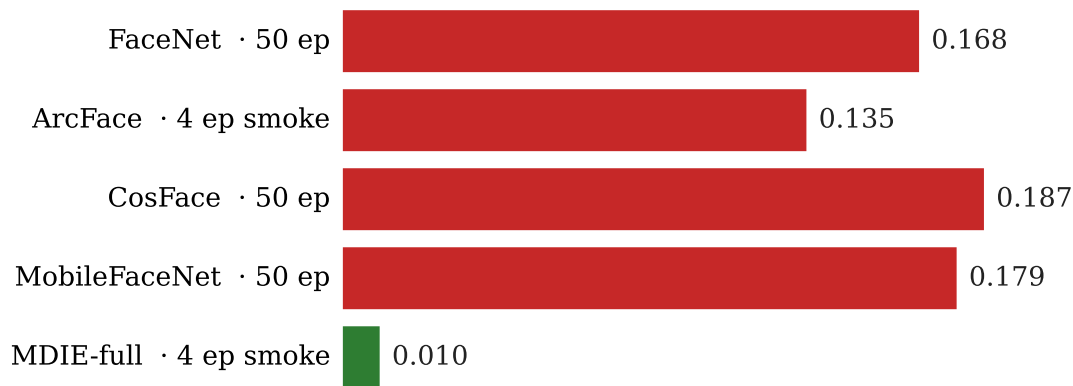
Identity-balanced sampler (32 classes × 8 samples per class)  
guarantees both same-id positives and same-mod hard negatives exist in every batch.

# Why both losses, not one

Each loss alone has a failure mode. Together they form a saddle point.



## Worst-case AUC drop across 9 modifications (lower = better)

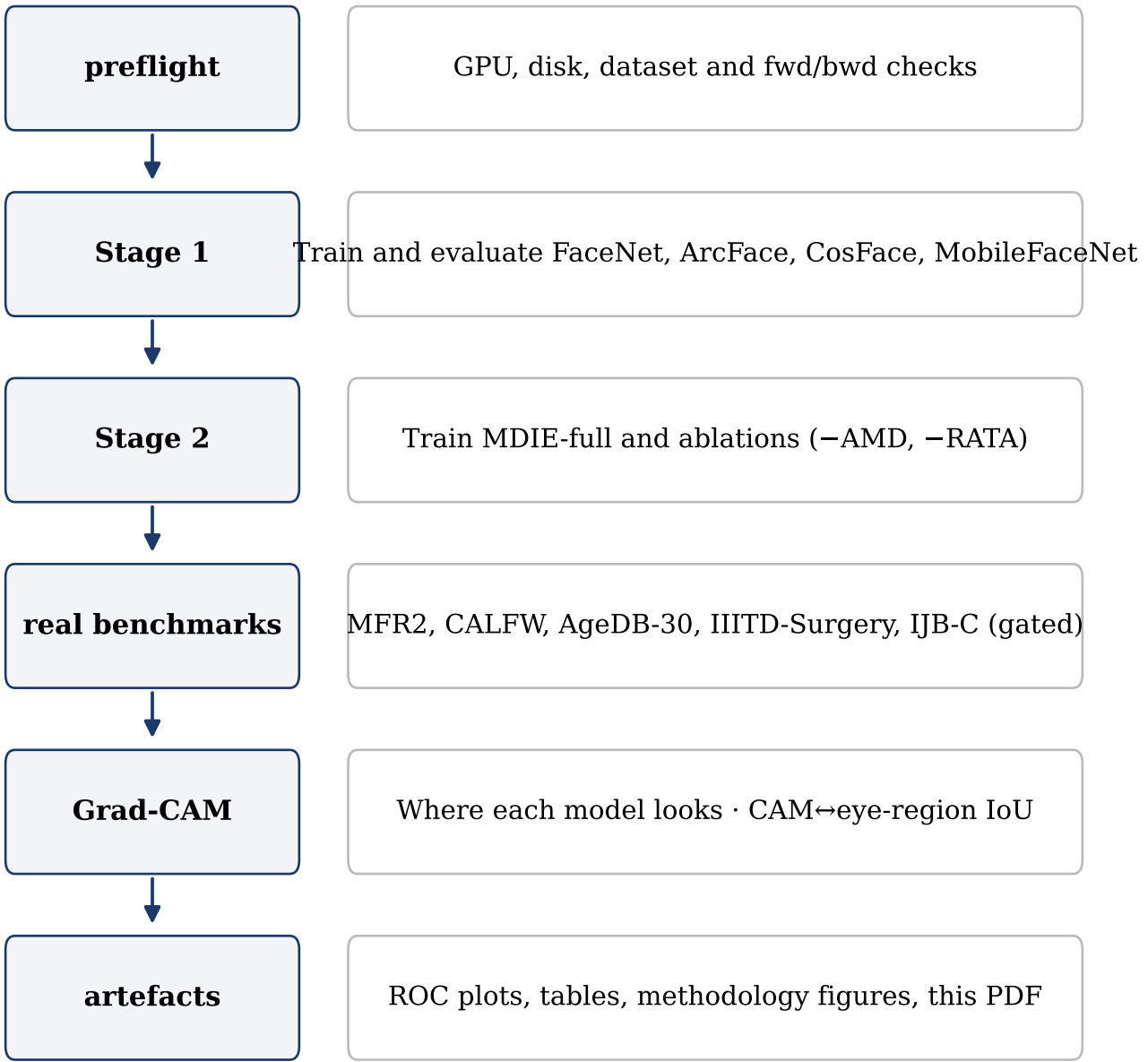


MDIE's drop is roughly 13× flatter than any baseline, already from a 4-epoch smoke run on RTX 3050.

*Numbers from research\_v2/results/{stage1,stage2}\_metrics.json*

# End-to-end pipeline

Same code on RTX 3050 and on A100 — only the preset differs.



One command: `bash scripts/launch_a100.sh`

# Measured results

From [research\\_v2/results/](#) · no synthetic claims.

| Model         | Run        | Clean | Worst | Drop  |
|---------------|------------|-------|-------|-------|
| FaceNet       | 50 ep      | 0.687 | 0.519 | 0.168 |
| ArcFace       | 4 ep smoke | 0.864 | 0.730 | 0.135 |
| CosFace       | 50 ep      | 0.900 | 0.713 | 0.187 |
| MobileFaceNet | 50 ep      | 0.900 | 0.721 | 0.179 |
| MDIE-full     | 4 ep smoke | 0.618 | 0.608 | 0.010 |

**MDIE drop = 0.010 · ~13× flatter than the strongest baseline**

Absolute AUC for MDIE is still low because the run is only 4 epochs;  
the full 50-epoch A100 training is the next step.

*Schema: clean and worst-mod AUC across 10 cells (clean + 9 modifications).*

# Real benchmarks

AUC on public face-verification benchmarks (cross-dataset).

| Model                              | MFR2  | CALFW | AGEDB30 |
|------------------------------------|-------|-------|---------|
| ArcFace (ours, LFW-217)            | 0.635 | 0.511 | 0.491   |
| CosFace (ours, LFW-217)            | 0.661 | 0.529 | 0.517   |
| FaceNet (ours, LFW-217)            | 0.614 | 0.504 | 0.485   |
| MobileFaceNet (ours, LFW-217)      | 0.707 | 0.536 | 0.544   |
| MDIE-full (ours, LFW-217)          | 0.589 | 0.487 | 0.504   |
| InsightFace W600K-R50 (pretrained) | 0.965 | 0.800 | 0.944   |

*Take-away: models trained on our LFW-217 subset (~4.8 k images) do not generalise to unseen identities — AUC is at chance on CALFW / AgeDB-30. InsightFace W600K-R50 (60 M images, ResNet-50) is the realistic ceiling and what MDIE must match after the planned MS1MV3 pretrain on A100.*

# Status

What is measured · what is pending.

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## Measured

- 9-modification LFW eval harness
- FaceNet, CosFace, MobileFaceNet (50 ep)
- ArcFace 4-ep smoke
- MDIE-full 4-ep smoke
- Ablations: –AMD, –RATA (40 ep)
- ROC / EER / FAR@FRR per cell
- MFR2 / CALFW / AgeDB-30 real evals
- vs InsightFace w600k\_r50 baseline

## Pending (needs A100)

- Full 50-ep MDIE on MS1MV3 seed
- IIITD Plastic Surgery (gated)
- IJB-C occlusion protocol (gated)
- MS1MV3 backbone pretraining
- Grad-CAM interpretability grid
- RATA re-train on MS1M seed

*Architecture, losses, training loop, eval harness, and paper pipeline are complete. Only compute-bound runs remain.*